Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102

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Digital Electronics: Course Goals and Key Takeaways

Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	**Combinational Logic Circuit:** Adder, Subtractor, Magnitude Comparator, Code Converters etc. **MSI Circuits:** Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	**Sequential Circuits:** Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:

1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.

Reference Books:

1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015.
2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Digital Electronics: Lecture Plan

Sl. No.	Topics	No of classes (tentative)
1.	Introduction to Number Systems and Conversion Techniques	8
2.	Continue…..	
3.	1's and 2's complement	
4.	Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR)	
5.	Boolean Algebra: Laws and Theorems	
6.	Boolean Function Implementation Using NAND/NOR Gates	
7.	Determination of Boolean Functions from Digital Circuits	
8.	Continue…..	
9.	SOP Standard Forms of Logical Functions	8
10.	POS Standard Forms of Logical Functions	
11.	Simplification of Logical Functions Using Boolean Algebra	
12.	Introduction to Karnaugh Maps (K-Maps)	
13.	K-Map Simplification (2-variable Functions)	
14.	K-Map Simplification (3-variable Functions)	
15.	K-Map Simplification (4-variable Functions)	
16.	Don't Care Conditions in K-Map Simplification	

Digital Electronics: Lecture Plan

17.	Introduction to Combinational Logic Circuits	8
18.	Design of Adders (Half Adder, Full Adder)	
19.	Design of Subtractors (Half Subtractor, Full Subtractor)	
20.	Magnitude Comparator Design	
21.	Code Converters (Binary to Gray, Gray to Binary, BCD to Excess-3)	
22.	Multiplexers: Design and Applications	
23.	Demultiplexers: Design and Applications	
24.	Decoders and Encoders	
25.	Introduction to Sequential Circuits	10
26.	Latches: SR Latch, D Latch	
27.	Flip-Flops: SR, JK Flip-Flops	
28.	Flip-Flops: D and T Flip-Flops	
29.	Asynchronous Counters: Design and Operation	
30.	Continue…..	
31.	Synchronous Counters: Design and Operation	
32.	Continue…..	
33.	Shift Registers: Types and Applications	
34.	Continue…..	

Digital Electronics: Weightage

Weightage of Different Components

1.	Quiz/ Assignment	10 Marks
2.	Mid Semester Examination	30 Marks
3.	Quiz/ Assignment	10 Marks
4.	End Semester Examination	50 Marks
Total Marks = 100		

K-Map

Example: Reduce the function $F(x,y,z) = \pi\,(0,1,2,3,5,7)$

Example: Reduce the function $F(A,B,C) = \pi\,(0,1,6,7)$

Don’t Care Condition in K-Map Simplification

- So far, the expressions considered have been completely specified for every combination of the input variables, that is, each minterm (maxterm) has been specified as a 1 or a 0.

- It often occurs that for certain input combinations, the value of the output is unspecified either because the input combinations are invalid or because the precise value of the output is of no consequence.

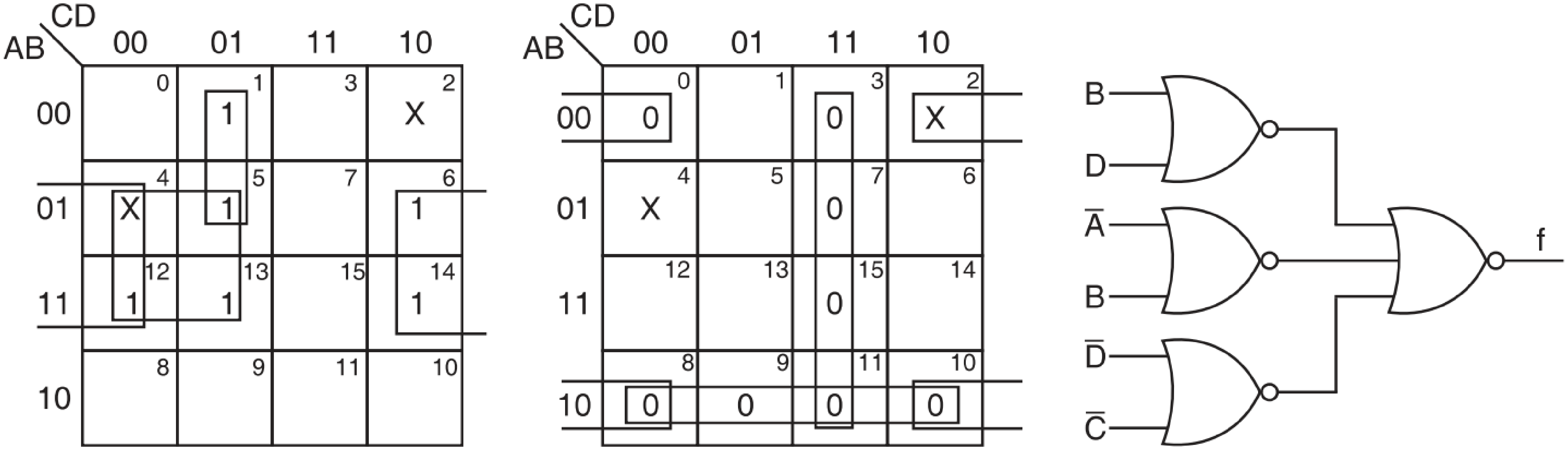
- The combinations for which the values of the expression are not specified are called *don’t care* combinations or *optional* combinations and such expressions, therefore, stand incompletely specified. The output is a don’t care for these invalid combinations.

- The don’t care terms are denoted by d, X. During the process of design using an SOP map, each don’t care is treated as a 1 if it is helpful in map reduction; otherwise, it is treated as a 0 and left alone.

- During the process of design using a POS map, each don’t care is treated as a 0 if it is useful in map reduction, otherwise it is treated as a 1 and left alone.

Don’t Care Condition in K-Map Simplification

Example: Reduce the expression $f = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$ and implement the real minimal expression in universal logic.



$f_{min} = B\overline{C} + B\overline{D} + \overline{A}\,\overline{C}D$

(a) SOP K-map

$f_{min} = (B + D)(\overline{A} + B)(\overline{C} + \overline{D})$

(b) POS K-map

(c) NOR logic

SOP minimal is

$$f_{min} = B\overline{C} + B\overline{D} + \overline{A}\,\overline{C}D$$

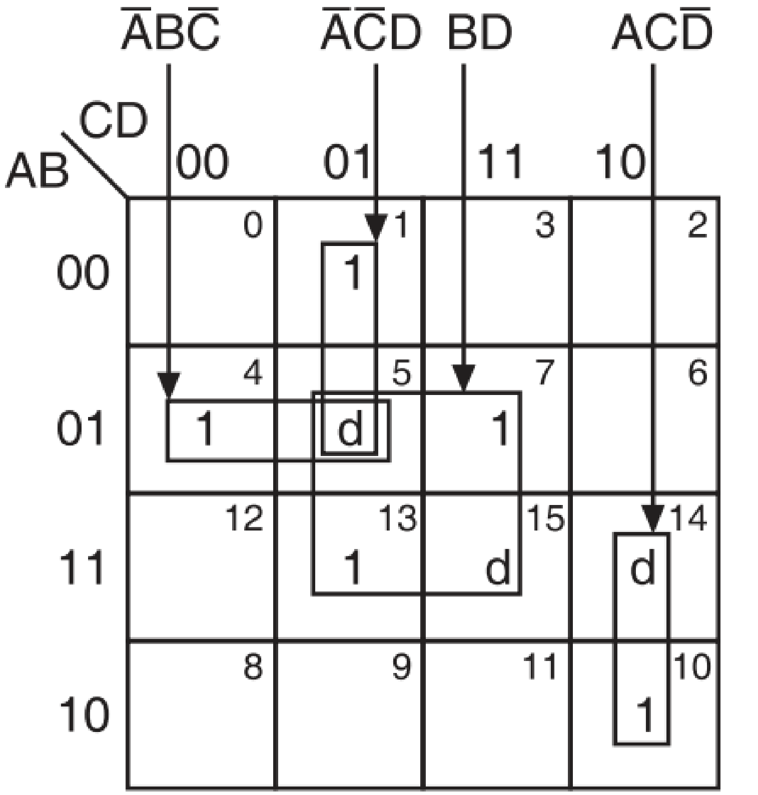
POS minimal is

$$f_{min} = (B + D)(\overline{A} + B)(\overline{C} + \overline{D}) = \overline{\overline{(B + D)} + \overline{(\overline{A} + B)} + \overline{(\overline{C} + \overline{D})}}$$

Don’t Care Condition in K-Map Simplification

Example: Minimize the following expressions using K-map:

(a) $F(A, B, C, D) = \Sigma\, m(1, 4, 7, 10, 13) + \Sigma\, d(5, 14, 15)$



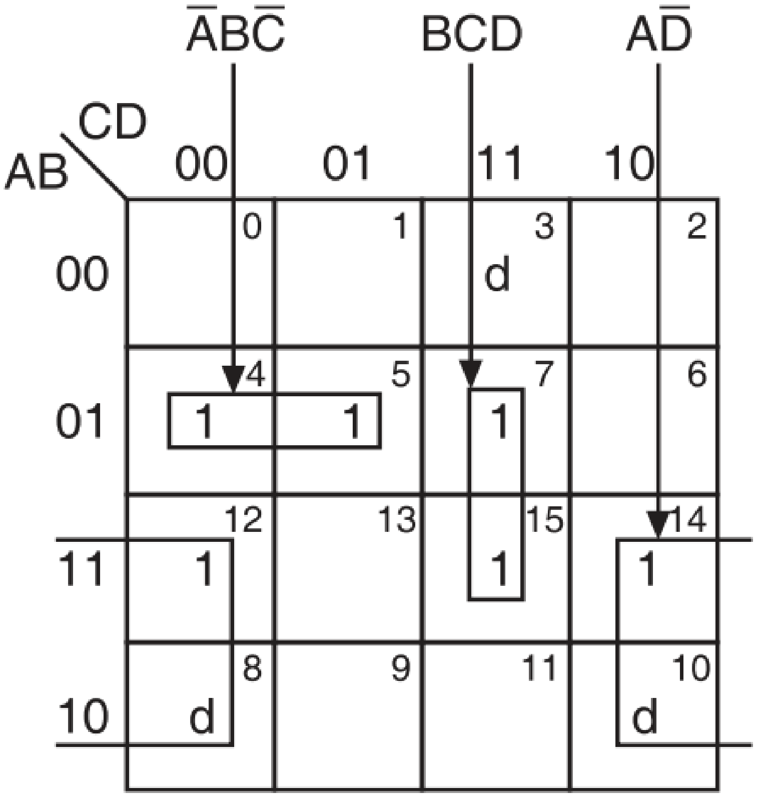
$f_{min} = \bar{A}B\bar{C} + \bar{A}\bar{C}D + BD + AC\bar{D}$

(a) SOP K-map

Don’t Care Condition in K-Map Simplification

Example: Minimize the following expressions using K-map:

(b) $F(A, B, C, D) = \Sigma\, m(4, 5, 7, 12, 14, 15) + \Sigma\, d(3, 8, 10)$



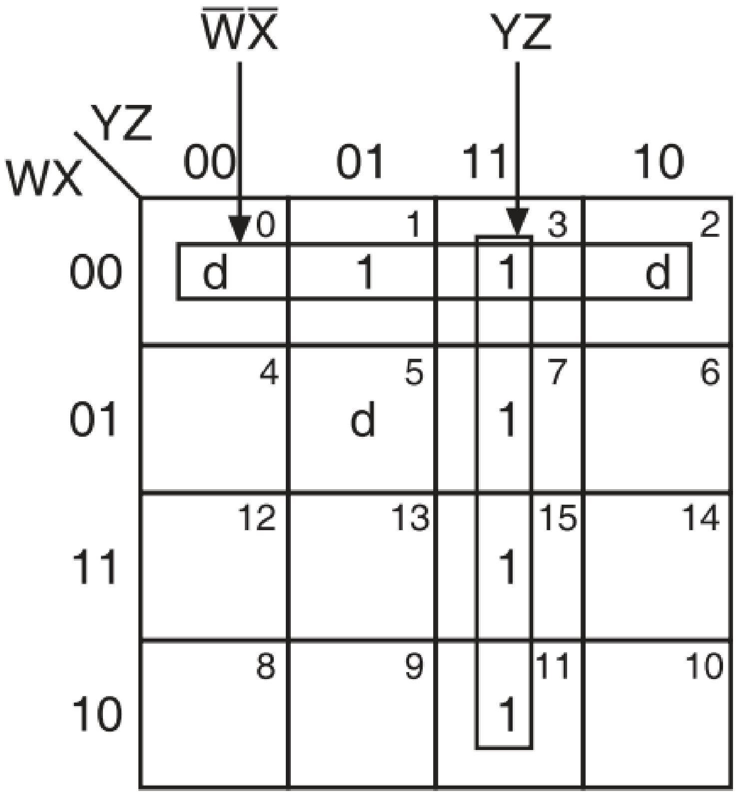
$f_{min} = A\bar{D} + BCD + \bar{A}B\bar{C}$

(b) SOP K-map

Don’t Care Condition in K-Map Simplification

Example: Minimize the following expressions using K-map:

(c) $F(W, X, Y, Z) = \Sigma m(1, 3, 7, 11, 15) + \Sigma d(0, 2, 5)$



$f_{min} = \overline{W}\overline{X} + YZ$

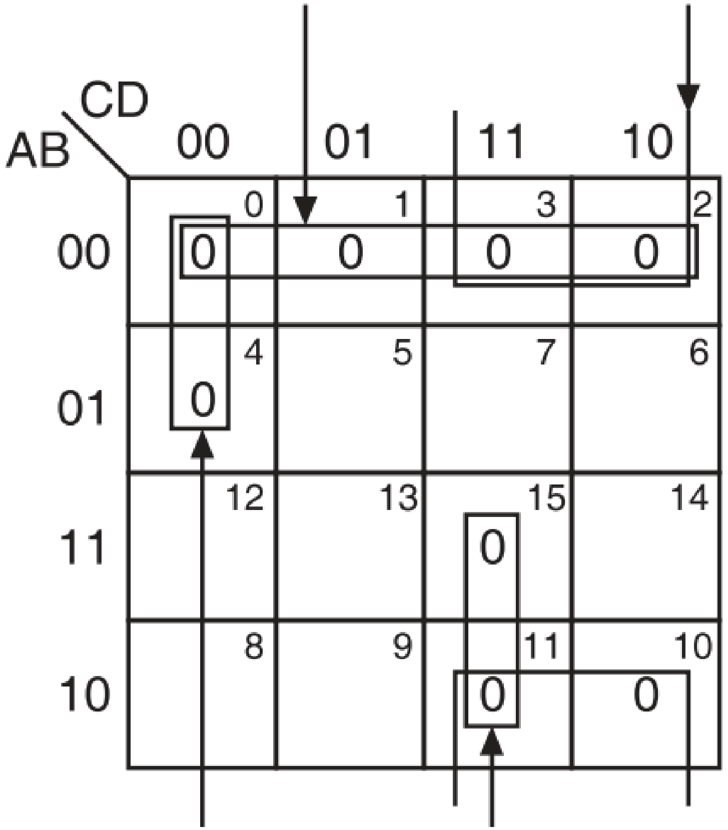
(c) SOP K-map

Don't Care Condition in K-Map Simplification

Example: Obtain the simplified expression in POS form

(a) $F(A, B, C, D) = \Sigma m(5, 6, 7, 8, 9, 12, 13, 14)$

(a) $F(A, B, C, D) = \Pi M(0, 1, 2, 3, 4, 10, 11, 15)$



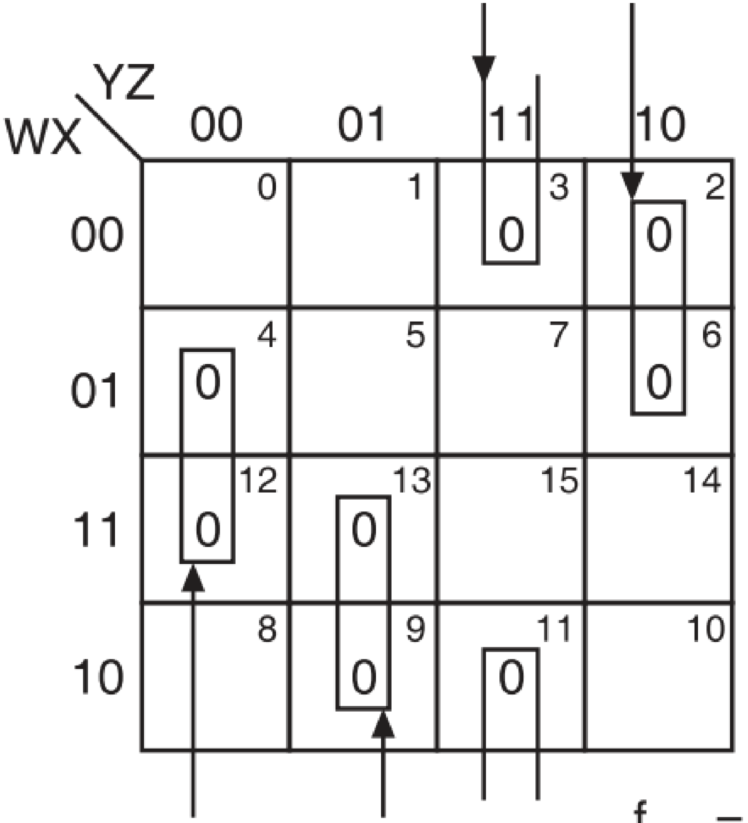
$f_{min} = (B + \bar{C})(A + B)(A + C + D)(\bar{A} + \bar{B} + \bar{C})$

Don’t Care Condition in K-Map Simplification

Example: Obtain the simplified expression in POS form

(b) $F(W, X, Y, Z) = \Sigma m(0, 1, 5, 7, 8, 10, 14, 15)$

(b) $F(W, X, Y, Z) = \Pi M(2, 3, 4, 6, 9, 11, 12, 13)$



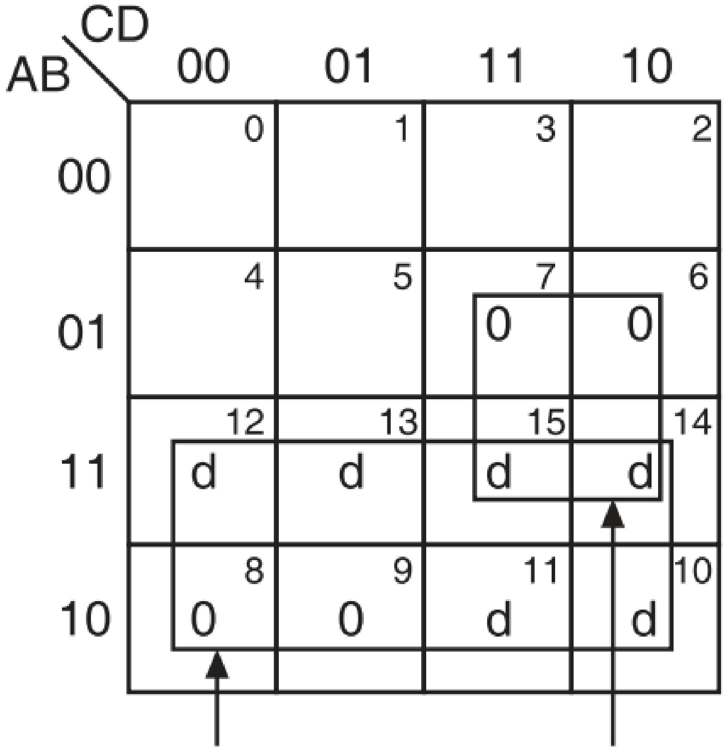
$f_{min} = (W + \bar{Y} + Z)(X + \bar{Y} + \bar{Z})(\bar{W} + Y + \bar{Z})(\bar{X} + Y + Z)$

Don’t Care Condition in K-Map Simplification

Example: Obtain the simplified expression in POS form

(c) $F(A, B, C, D) = \Sigma\, m(0, 1, 2, 3, 4, 5) + d(10, 11, 12, 13, 14, 15)$

(c) $F(A, B, C, D) = \Pi\, M(6, 7, 8, 9).\, d\, (10, 11, 12, 13, 14, 15)$

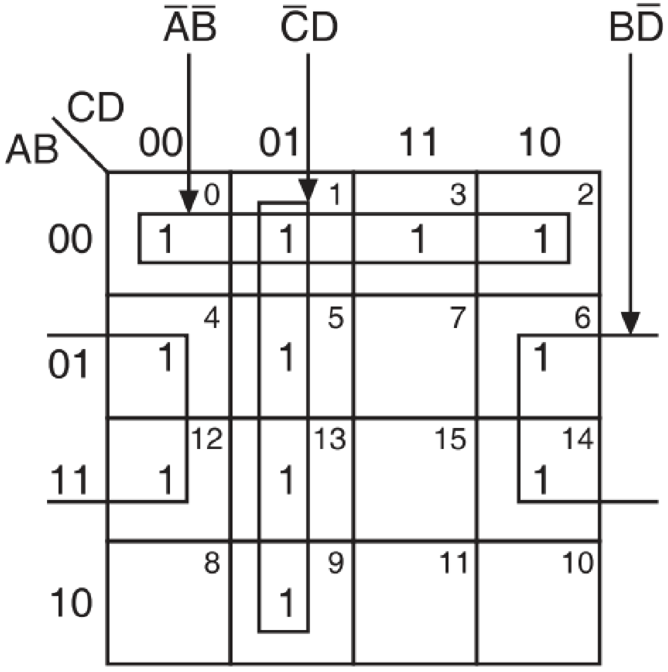


$f_{min} = \bar{A}(\bar{B} + \bar{C})$

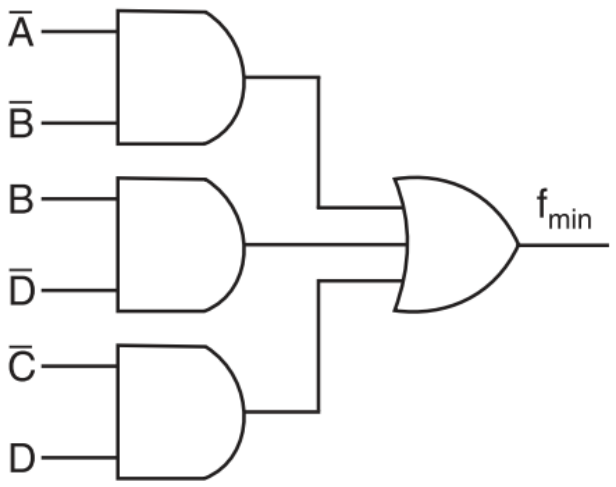
Example: (a) Design a logic circuit using minimum number of basic gates for the

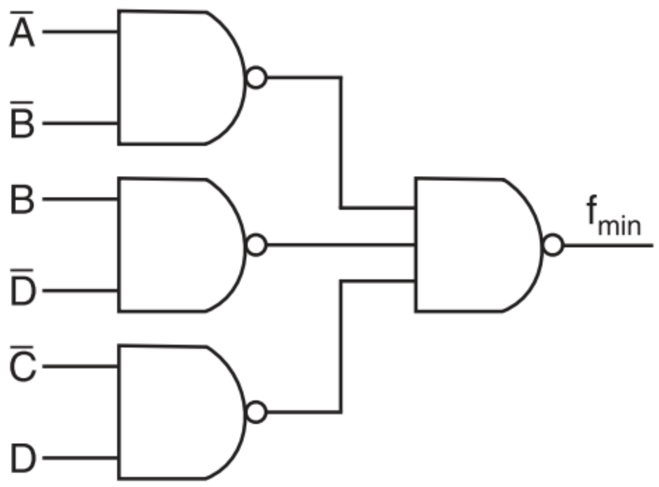
following Boolean expression $f = (\overline{A}\,\overline{B}\,\overline{C}\,\overline{D}) + \overline{A}\,\overline{B}\,\overline{C}D + (\overline{A}\,\overline{B}C\overline{D}) + (\overline{A}\,\overline{B}CD) + (\overline{A}B\overline{C}\,\overline{D}) + (\overline{A}B\overline{C}D) + (\overline{A}BC\overline{D}) + (A\overline{B}\,\overline{C}D) + (AB\overline{C}\,\overline{D}) + (AB\overline{C}D) + (ABC\overline{D})$.

$$f = \Sigma\, m(0, 1, 2, 3, 4, 5, 6, 9, 12, 13, 14)$$



$f_{min} = \overline{A}\overline{B} + \overline{C}D + B\overline{D}$





Don’t Care Condition in K-Map Simplification

Example: Simplify the Boolean expression using K-map $F = \bar{A} + AB + AB\bar{D} + A\bar{B}\bar{D} + C$

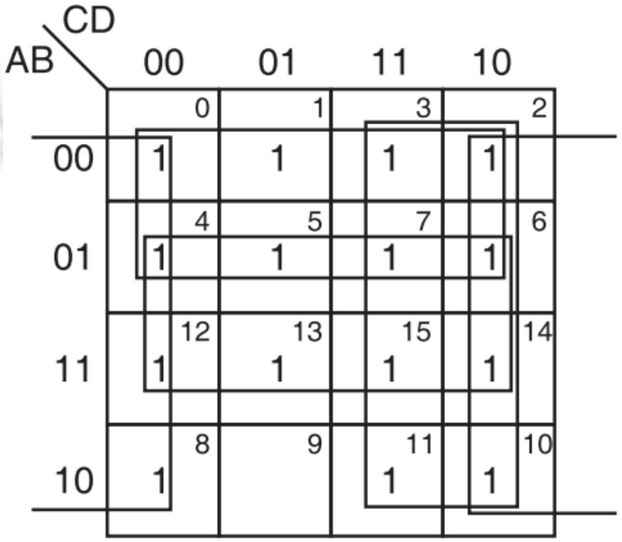
$$F = \bar{A} + AB + AB\bar{D} + A\bar{B}\bar{D} + C = 0XXX + 11XX + 11X0 + 10X0 + XX1X$$

So the minterms are: 0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1100, 1101,1110, 1111, 1100, 1110, 1000, 1010, 0010, 0011, 0110, 0111, 1010, 1011, 1110, 1111.

So

$$F = \Sigma\, m(0, 1, 2, 3, 4, 5, 6, 7, 12, 13, 14, 15, 12, 14, 8, 10, 2, 3, 6, 7, 10, 11, 14, 15)$$

$$= \Sigma\, m(0, 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15)$$



$f_{min} = \bar{A} + B + C + \bar{D}$

(a) SOP K-map

Don’t Care Condition in K-Map Simplification

Example:

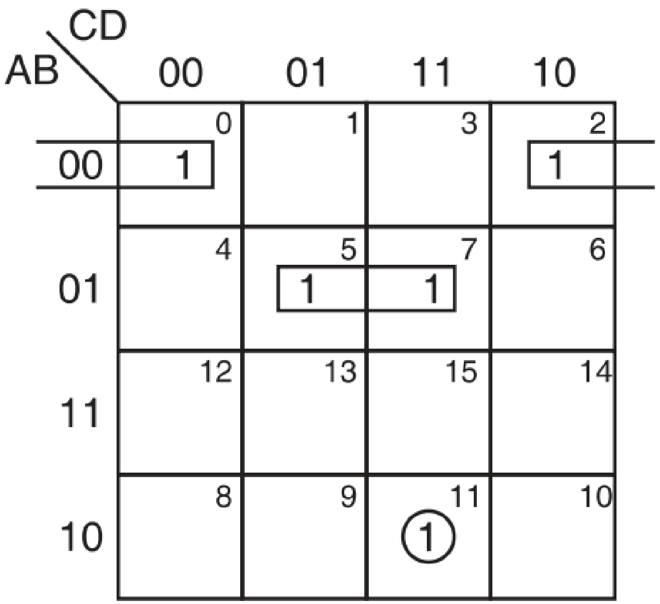
(b) Obtain the simplified expression using K-map $F = \bar{A}\bar{B}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}BD + AB\bar{C}D$.

$$F = \bar{A}\bar{B}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}BD + AB\bar{C}D = 00X0 + 0000 + 01X1 + 1101$$

So the minterms are: 0000, 0010, 0000, 0101, 0111, 1101.

So

$$F = \Sigma\, m(0, 2, 5, 7, 11)$$



$f_{min} = \bar{A}\bar{B}\bar{D} + \bar{A}BD + A\bar{B}CD$

(b) SOP K-map

Don't Care Condition in K-Map Simplification

Example:

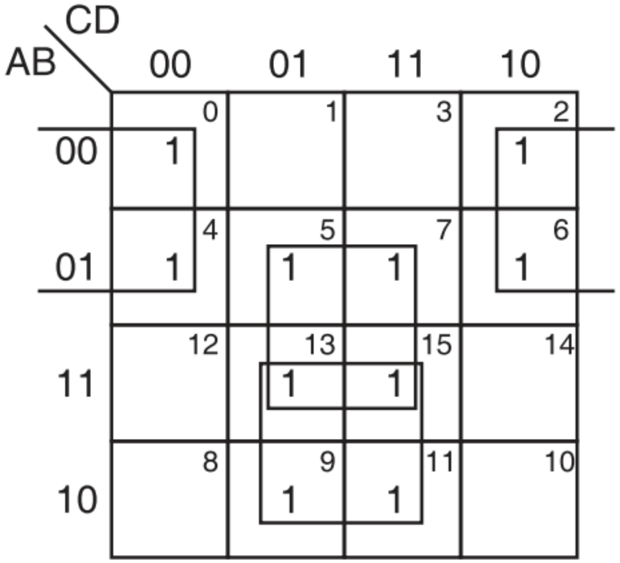
(c) Obtain the simplified expression using K-map $F = ABD + \bar{A}\bar{C}\bar{D} + \bar{A}B + \bar{A}C\bar{D} + A\bar{B}D$.

$$F = ABD + \bar{A}\bar{C}\bar{D} + \bar{A}B + \bar{A}C\bar{D} + A\bar{B}D = 11X1 + 0X00 + 01XX + 0X10 + 10X1$$

So the minterms are: 1101, 1111, 0000, 0100, 0100, 0101, 0110, 0111, 0010, 0110, 1001, 1011.

So

$$F = \Sigma\, m(13, 15, 0, 4, 4, 5, 6, 7, 2, 6, 9, 11) = \Sigma\, m(0, 2, 4, 5, 6, 7, 9, 11, 13, 15)$$



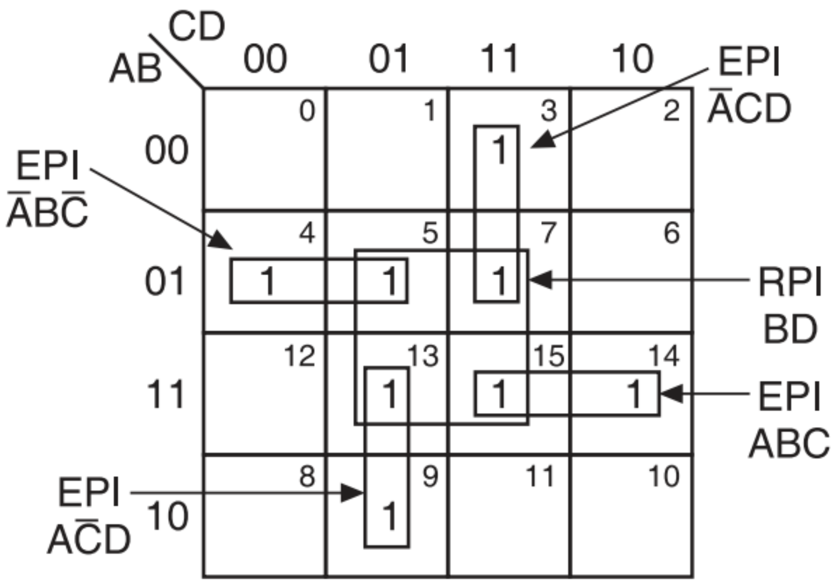
$f_{min} = \bar{A}\bar{D} + BD + AD$

(c) SOP K-map

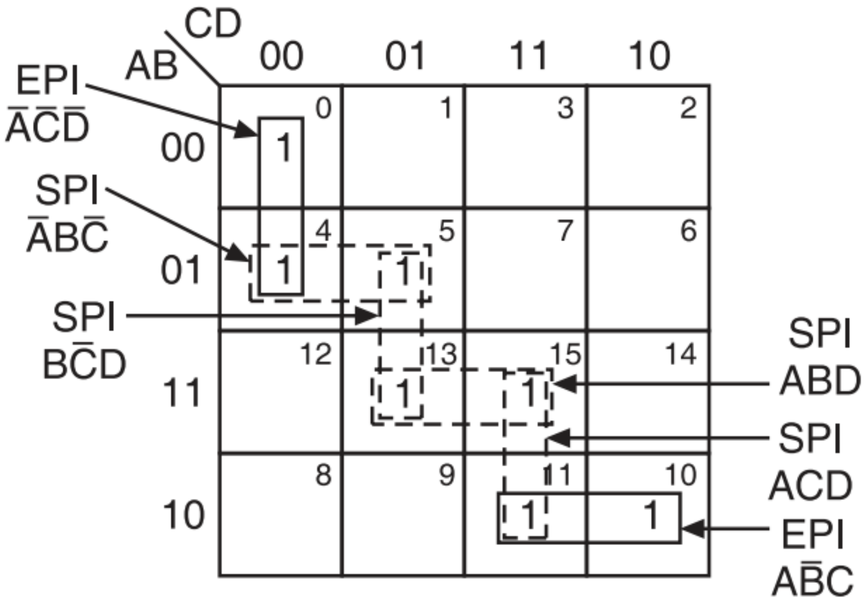
Prime Implicants and Essential Prime Implicants in K-Map

- Each square or rectangle made up of the bunch of adjacent minterms is called a subcube.
- Each of these sub cubes is called a prime implicant (PI).
- The prime implicant which contains at least one 1 which cannot be covered by any other prime implicant is called an essential prime implicant (EPI).
- The prime implicant whose each 1 is covered at least by one EPI is called a redundant prime implicant (RPI).
- A prime implicant which is neither an essential prime implicant nor a redundant prime implicant is called a selective prime implicant (SPI).

$$F(A, B, C, D) = \bar{A}CD + ABC + A\bar{C}D + \bar{A}B\bar{C}$$

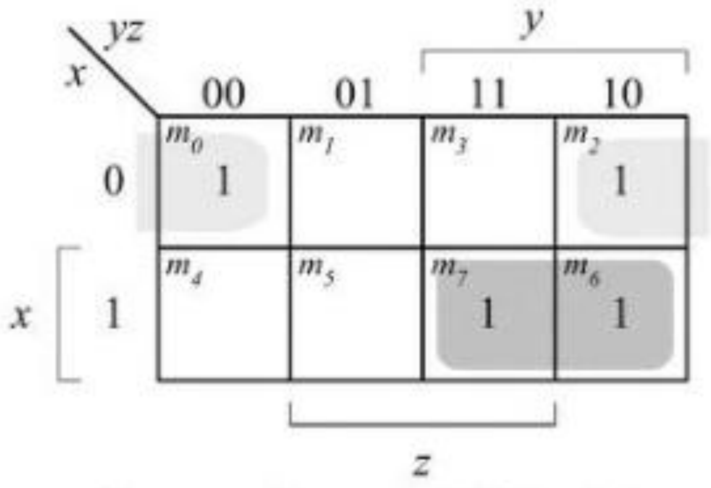


$$F(A, B, C, D) = \Sigma\, m(0, 4, 5, 10, 11, 13, 15)$$



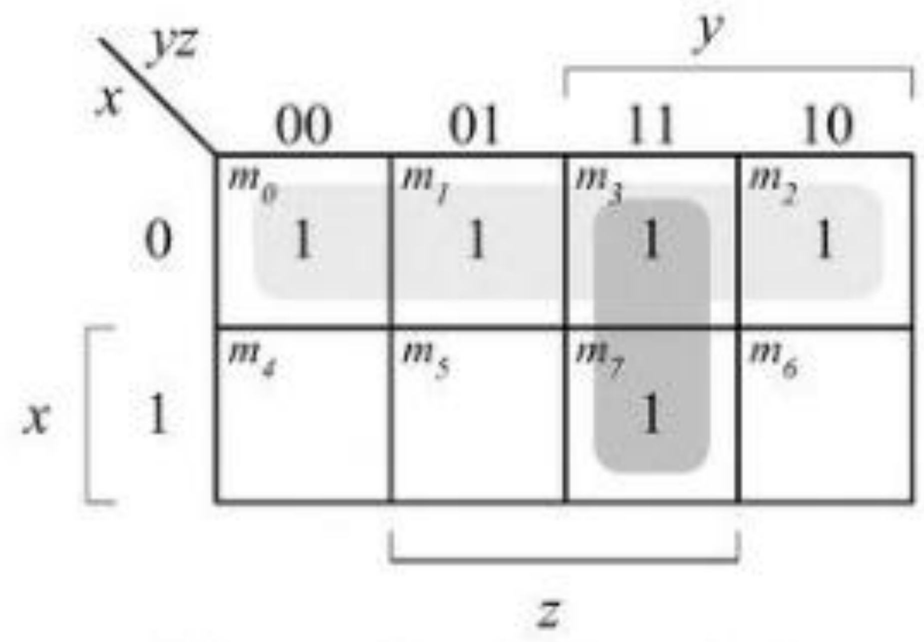
Problems

Problem: Simplify the following Boolean function using three variable K-Map:
$$F(x, y, z) = xy + \bar{x}\,\bar{y}\,\bar{z} + \bar{x}\,y\,\bar{z}$$



(a) $F = xy + x'y'z' + x'yz'$
$F = xy + x'z'$

Problem: Simplify the following Boolean function using three variable K-Map:
$$F(x, y, z) = \bar{x}\,\bar{y} + yz + \bar{x}\,y\,\bar{z}$$

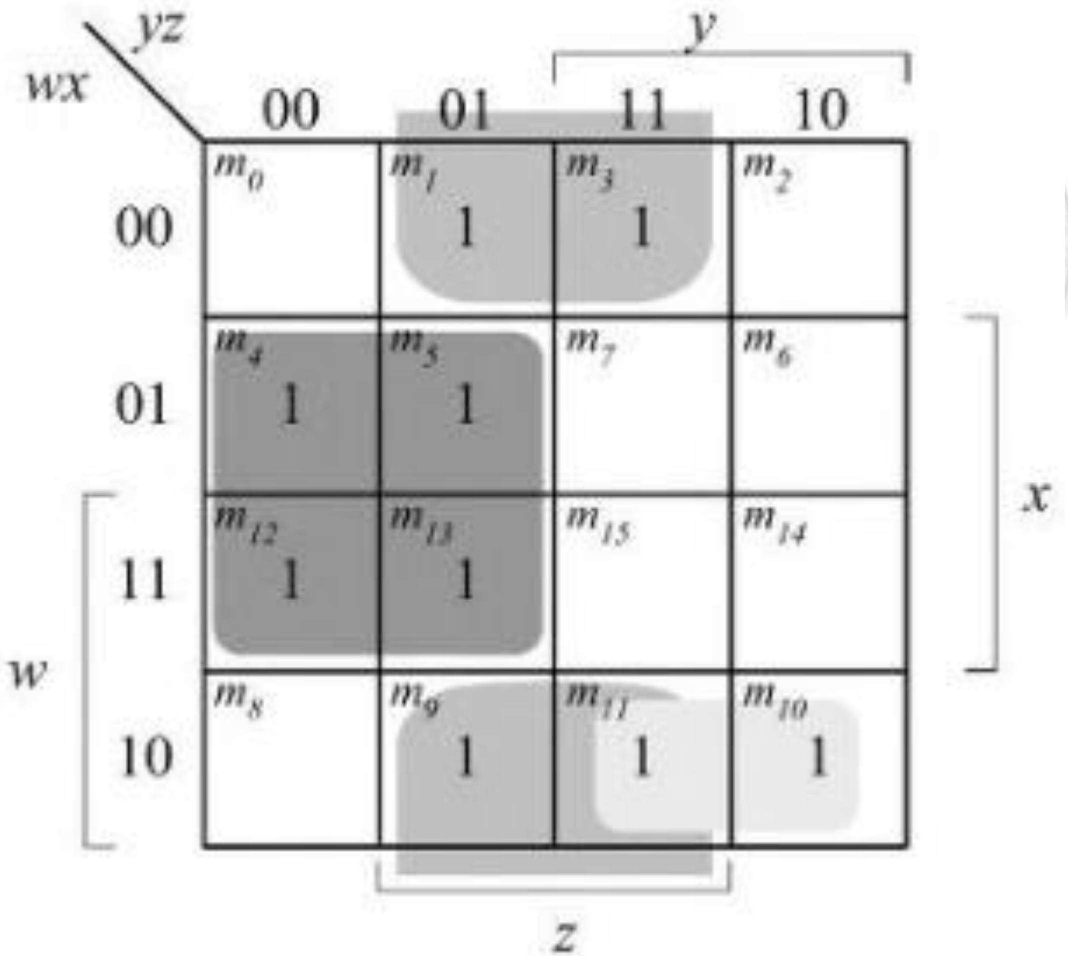


(b) $F = x'y' + yz + x'yz'$
$F = x' + yz$

Problems

Problem: Simplify the following Boolean function using four variable K-Map:

$$\bar{x}\,z + \bar{w}\,x\,\bar{y} + w(\bar{x}\,y + x\bar{y})$$

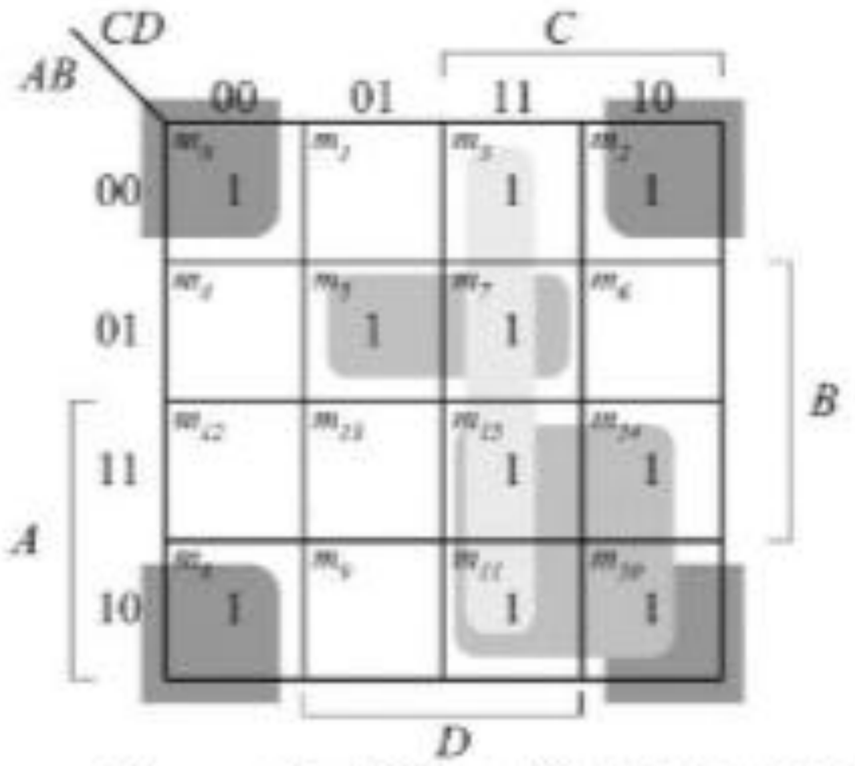


(b) $F = xy' + x'z + wx'y$

Problems

Problem: Simplify the following Boolean function using four variable K-Map:

$$A\,\overline{B}\,C + \overline{B}\,\overline{C}\,\overline{D} + BCD + AC\overline{D} + \overline{A}\,\overline{B}C + \overline{A}B\overline{C}D$$



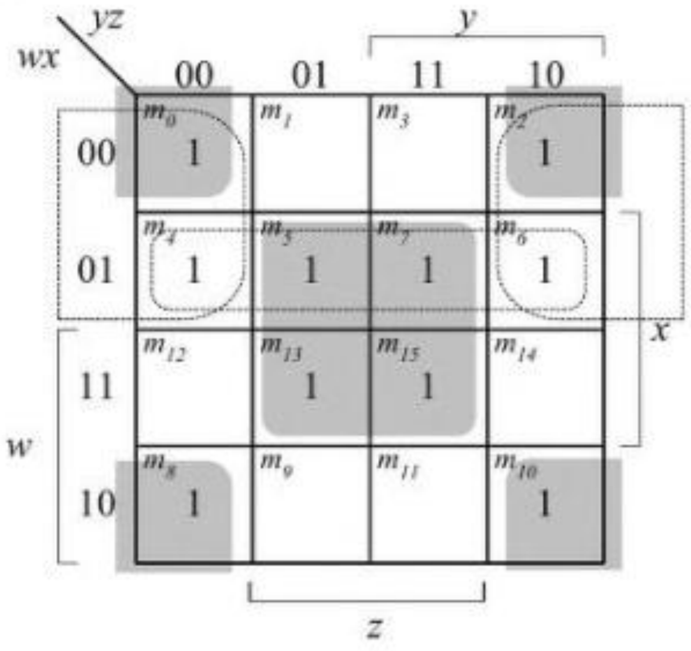
(c) $F = B'D' + AC + A'BD + CD$ (or $B'C$)

Problems

Problem: Find all the prime implicants for the following Boolean functions and determine which are essential:

$$F(w, x, y, z) = \sum(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$$

(a)



Essential: xz, $x'z'$

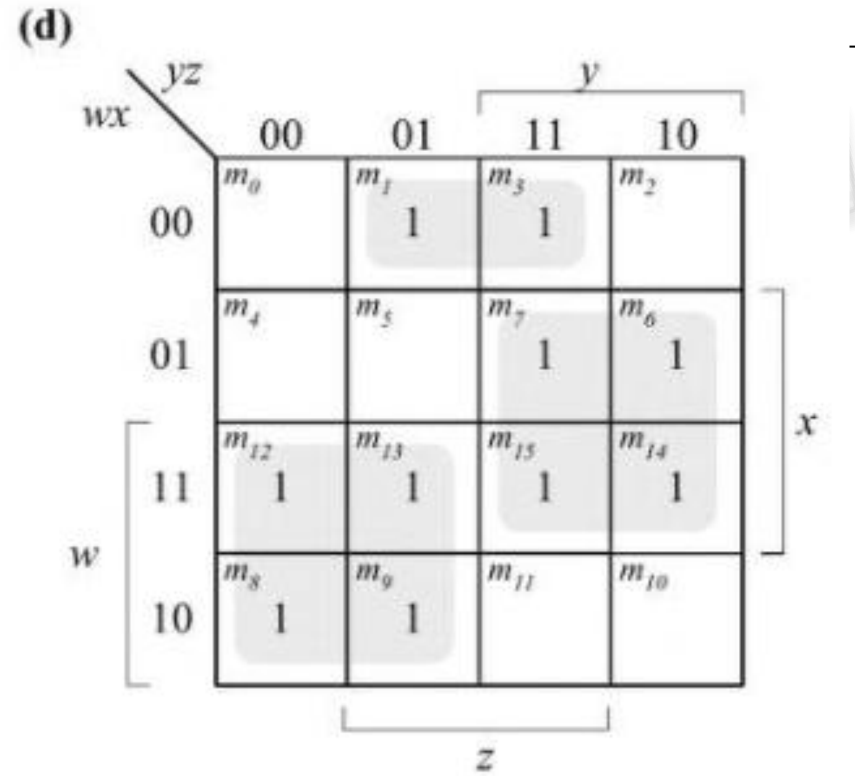
Non-essential: $w'x$, $w'z'$

$F = xz + x'z' + (w'x \text{ or } w'z')$

Problems

Problem: Find all the prime implicants for the following Boolean functions and determine which are essential:

$$F(w,x,y,z) = \sum(1,3,6,7,8,9,12,13,14,15)$$



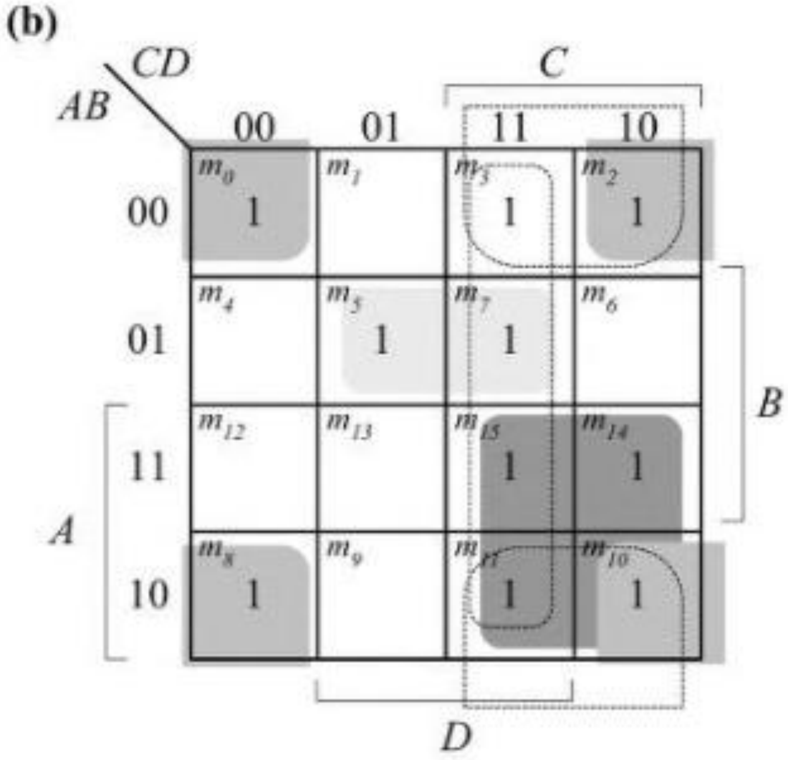
Essential: wy', xy, $w'x'z$

$F = wy' + xy + w'x'z$

Problems

Problem: Find all the prime implicants for the following Boolean functions and determine which are essential:

$$F(A,B,C,D) = \sum(0,2,3,5,7,8,10,11,14,15)$$

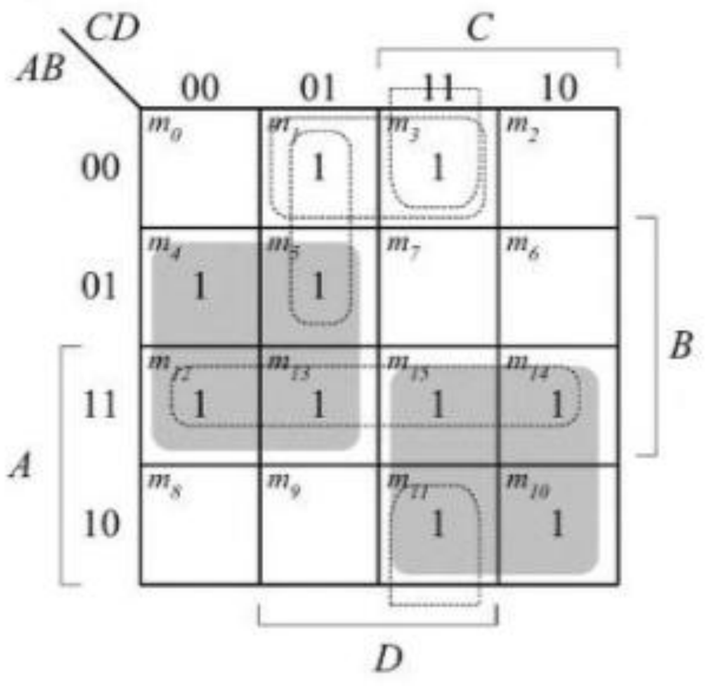


Essential: *B'D', AC, A'BD*
Non-essential: *CD, B'C*
F = B'D' + AC + A'BD + (CD OR B'C)

Problems

Problem: Find all the prime implicants for the following Boolean functions and determine which are essential:

$$F(A,B,C,D) = \sum(1,3,4,5,10,11,12,13,14,15)$$



Essential: BC', AC

Non-essential: *AB, A'B'D, B'CD, A'C'D*

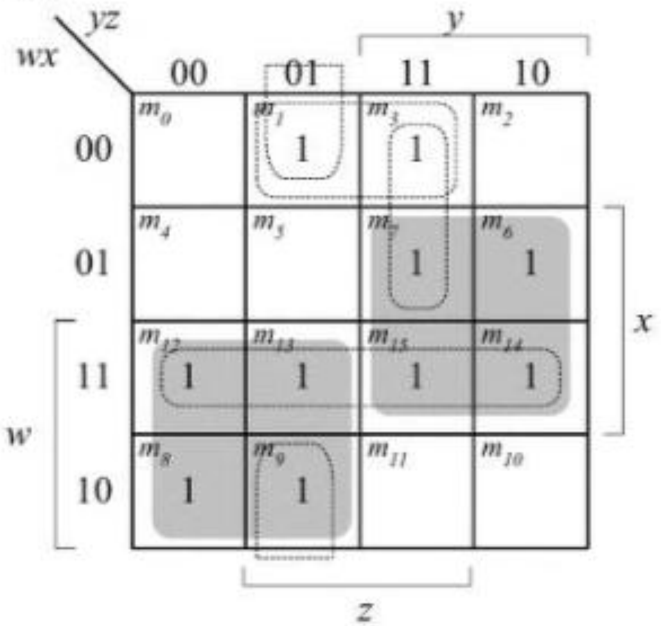
F = BC' + AC + A'B'D

Problems

Problem: Find all the prime implicants for the following Boolean functions and determine which are essential:

$$F(w,x,y,z) = \sum(1,3,6,7,8,9,12,13,14,15)$$

(d)



Essential: wy', xy

Non-essential: wx, $x'y'z$, $w'wz$, $w'x'z$

$F = wy' + xy + w'x'z$